Neural Network Surrogates for TGLF: Prediction Accuracy and Uncertainty

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Abstract

Repeated evaluations of plasma transport models can be costly. This project investigates methods for approximating the ion heat, electron heat, and electron particle fluxes predicted by the trapped gyro-Landau fluid (TGLF) model. We explore the supplied simulation data, compare candidate models, and test choices in input representation, network design, and training. An initial study of 43 scan cases shows that greater model flexibility does not consistently improve prediction and that low aggregate error can coexist with incorrect flux signs. On a larger parameter-scan database, a five-network ensemble reduces test root mean squared error by 59.6%–77.2% relative to a linear baseline across the three outputs. A controlled activation comparison also reveals different tradeoffs for heat and particle flux prediction. These are preliminary results from an exploratory evaluation process. They support further development of the ensemble while motivating checks of physical trends, training data requirements, and uncertainty. A complete ranking of the group's candidate methods remains open.

Code: `https://github.com/lindbergryan04/dsc180a-project-1`

1 Introduction

1.1 Problem and project scope

Turbulence transports heat and particles across a tokamak plasma and affects its temperature and density profiles. Predicting these fluxes is part of modeling plasma confinement. Detailed turbulence simulations are expensive, and even reduced models such as TGLF can become costly when a larger calculation requires many evaluations (Meneghini et al. 2017; Cao et al. 2026). A surrogate learns to approximate a simulation from previously computed input-output pairs, making repeated predictions less expensive.

Our research question is which modeling approaches and design choices provide useful approximations of TGLF under the available data and computational constraints. We began by examining the plasma inputs, output scales, and data quality, then explored regression, tree-based, and neural network methods. We assess choices through prediction error, computational cost, and behavior across plasma conditions. Uncertainty estimates provide an additional way to examine model reliability. DIII-D is an application within this investigation.

The prediction targets are ion heat flux Q_i, electron heat flux Q_e, and electron particle flux Γ_e. The reference values are TGLF outputs. Agreement with these values measures approximation of TGLF; it does not by itself establish agreement with experimental measurements or nonlinear gyrokinetic simulations. This checkpoint reports an exploratory comparison on a small scan dataset and a more developed neural network study on a larger database. The distinction between these experiments matters when interpreting which methods performed well.

1.2 Literature review and prior work

Meneghini et al. (2017) developed neural network surrogates of TGLF and EPED1 and incorporated them into a coupled core and pedestal modeling workflow. Their TGLF architecture description in Section 3.1 specifies an ensemble of 32 independently initialized networks, each with one hidden layer of 60 neurons. This provides a precedent for averaging networks and a reference for investigating ensemble size. It does not establish an optimal member count for our data. Our deeper network architecture and training database differ from theirs, so our experiments evaluate the approach in a different setting.

Model evaluation also requires attention to local behavior. van de Plassche et al. (2020) developed QLKNN, a neural network surrogate of the QuaLiKiz transport model. Their work emphasizes evaluating local transport behavior alongside aggregate prediction error, motivating our checks of flux trends and individual prediction failures. Cao et al. (2026) developed TGLF-WINN using feature engineering, transformed flux targets, and supervision of flux contributions at individual wavenumbers. Their comparisons of individual design components illustrate how controlled experiments can identify useful modeling choices. We use integrated flux targets and test simpler representations and network choices; we do not implement their wavenumber supervision in the experiments reported here.

Training data selection is another direction in this literature. Zanisi et al. (2024) introduced Active Deep Ensembles for Plasma Turbulence, using neural network ensembles to guide training-data selection for QuaLiKiz surrogates. This provides context for studying data requirements and uncertainty, although our current experiments use fixed training datasets rather than active acquisition. Finally, Dudding et al. (2022) developed the SAT3 saturation rule using nonlinear CGYRO simulations, with particular attention to isotope dependence. The course's starter dataset draws on that work, but the added TGLF outputs supplied for our analysis use SAT2. The saturation rule and reference simulation must therefore remain explicit.

1.3 Relevant data

The starter dataset contains 43 plasma cases arranged in controlled scans around a reference condition. Case names identify 26 letter groups and an ion isotope. Eight groups contain complete hydrogen, deuterium, and tritium triplets. Some inputs change together: for example, the electron and ion temperature gradients are equal in these cases. These dependencies limit the separate effects that can be inferred from correlations. We retained all 43 cases after basic validity checks, including cases flagged for further review.

The larger `scan_db` dataset contains 5,000 TGLF cases with multiple inputs varied simultaneously over prescribed ranges. The sampled continuous coordinates use uniform distributions, with collision frequency sampled logarithmically. Unlike the starter scans, electron and ion temperature gradients vary independently. Each case supplies a complete input listing in `input.tglf.gen` and species-resolved fluxes in `out.tglf.gbflux`. We retain the supplied gyro-Bohm normalization and the sign of particle flux.

Additional supplied data support focused checks. The `scan_db_outside` collection contains nine groups of 30 cases, each placing one input outside its training range. The `data_d3d` collection includes 300 TGLF evaluations with perturbed inputs around a DIII-D reference condition. These serve as evaluation cases in the uncertainty study; they are distinct from the new DIII-D training database described in the remaining work.

2 Methods

2.1 Data preparation and representation

We parsed one row per case, checking numeric values, species identity, and flux-file length before assigning output labels. In `scan_db`, we fixed a 70%/15%/15% training, validation, and test split using random seed 42. Four cases with negative ion heat flux were held for review; all four had been assigned to training. Their signs agree with the labeled run output, so the exclusions do not reflect a simple flux-column parsing error. The original rows and split assignments were preserved. This leaves 3,496 training, 750 validation, and 750 test cases. Negative particle flux alone was not an exclusion criterion.

The larger-dataset models use 11 inputs: separate temperature gradients, a shared density gradient, the ion-to-electron temperature ratio, safety factor, elongation, flux-surface shift, normalized minor radius, ion mass, logarithmic collision frequency, and magnetic shear. We removed constant and duplicate coordinates. Magnetic shear was reconstructed from the TGLF inputs as

$$s = \frac{\mathtt{Q_PRIME_LOC}\,\mathtt{RMIN_LOC}^2}{\mathtt{Q_LOC}^2}.$$

Identifiers and other flux outputs were excluded from the predictors.

The positive heat targets span several orders of magnitude. We modeled $\log_{10} Q_i$ and $\log_{10} Q_e$, while retaining signed Γ_e on its original scale. Inputs and transformed targets

were standardized using training rows only. Heat predictions were converted back to flux units before evaluation. This transformation changes the fitting objective; improved fit in logarithmic space need not imply lower error in original units.

2.2 Exploratory models on the starter scans

The initial comparison used ten physical input coordinates on the 43 starter cases, representing the tied temperature gradients with one coordinate. We compared a training-mean baseline, linear regression variants, additive splines, quadratic terms with interactions, and a random forest. The variants tested logarithmic collision frequency and, for heat fluxes, logarithmic targets. The forest used 100 trees, maximum depth four, and a minimum leaf size of three.

For outer evaluation, we held out one entire letter group at a time, keeping related isotope cases together. Inner five-fold validation, also grouped by letter, selected regression ridge penalties from $\{0.1, 1, 10, 100\}$ using case-level mean absolute error (MAE). Preprocessing was fitted within the training folds. The forest and mean baseline had fixed settings. We report outer MAE averaged first within each letter and then across letters, giving each group equal weight. Model families were compared using these same outer scores, so this is an exploratory screen rather than an untouched final test.

2.3 Neural network development on the larger database

We compared a five-network ensemble with two baselines on `scan_db`. The mean baseline predicts each training target's arithmetic mean in flux units. The linear baseline uses standardized inputs and the same transformed heat and signed particle targets as the networks, without regularization or interaction terms. These baselines differ from the grouped ridge models in the starter-data experiment.

Each network has layer widths 11–64–64–8–3 and a linear output layer. The five members use seeds 42–46. Training uses AdamW, batch size 256, and weight decay 10^{-4}, with 30 warmup epochs at learning rate 0.001 and up to 120 further epochs at 0.003. Validation selects the saved checkpoint, with early stopping after 25 epochs without improvement. Each member's heat outputs are inverted before taking the equal-weight average of all five predictions.

We tested four activation combinations. The first activation was applied to both 64-unit layers, and the second to the 8-unit layer: GELU with tanh, ReLU with tanh, GELU with ReLU, and ReLU throughout. Across combinations, we matched initial weights and mini-batch order within each seed and held the architecture, data, scalers, and training settings fixed. The selection score was

$$S = \frac{1}{3} \sum_{j \in \{Q_i, Q_e, \Gamma_e\}} \frac{\mathrm{RMSE}_{\mathrm{val},j}}{\mathrm{SD}_{\mathrm{train},j}}, \tag{1}$$

with both quantities measured in original flux units. This prevents a large-scale target from dominating selection. The learning rate came from an earlier GELU/tanh search; the alternatives were not separately tuned. GELU/ReLU was selected, and its five checkpoints were retained unchanged.

This search was limited to four activation combinations on one architecture and one data split, using five matched seeds and a shared training schedule. The improvement in aggregate validation error is promising, although the selected configuration increased electron-heat error relative to the control. These findings motivate further tests across architectures and separately tuned training settings rather than a final choice of activation functions.

2.4 Evaluation and supplementary uncertainty checks

We report root mean squared error (RMSE) and R^2 in addition to the selection score. Learning curves use nested training subsets of 250, 500, 1,000, 2,000, and 3,496 cases, defined by a fixed permutation with seed 2026. Scalers were refitted on each subset, with the same validation set for checkpoint selection. We also evaluated the selected ensemble on all 43 starter cases and checked the directions of temperature-gradient and isotope trends. These cases were already used for exploration, but did not select the activation configuration.

The 750-case random test set had been inspected in earlier work. Consequently, the reported comparisons are exploratory and do not constitute a new, untouched assessment. The learning curves also use only one nested sequence. With a fixed maximum epoch budget, larger subsets receive more minibatch updates per epoch, so the curves do not isolate data volume from optimization effort.

A separate 15-network experiment assessed uncertainty using member disagreement. For each flux, prediction intervals were centered on the ensemble mean. Their half-width was a calibrated multiplier times the larger of the member sample standard deviation and 1% of the training-target standard deviation. Multipliers were fitted on the 750 validation cases for nominal 90% coverage, then fixed for subsequent checks. Because validation had already supported model selection, coverage is an empirical assessment rather than a formal guarantee. These results are reported separately from the five-network activation comparison and do not measure the effect of increasing ensemble size.

3 Preliminary results

3.1 What the small-data comparisons reveal

Table 1 shows that greater flexibility did not consistently improve the starter-data predictions. Replacing raw collision frequency with its logarithm improved linear-model MAE for all three targets. The fixed forest performed worse than that transformed linear model on each target. The logarithmic heat model had the lowest electron-heat MAE, while quadratic terms had the lowest ion-heat MAE.

Table 1: Exploratory starter-data comparison: letter-balanced MAE in gyro-Bohm units across 26 held-out letter groups. Lower is better. These scores use a different dataset and procedure from Table 2. The logarithmic heat model was not applied to particle flux.

Model	Q_i	Q_e	Γ_e
Training mean	17.022	14.830	6.127
Raw linear	9.570	10.215	3.520
Log-collision linear	8.295	7.695	2.725
Additive splines	17.528	11.475	3.911
Quadratic + interactions	7.965	7.253	3.210
Random forest	16.302	11.267	3.484
Log-heat linear	10.213	5.306	—

The lowest MAE did not ensure suitable behavior in every case. The quadratic ion-heat model predicted -1.34 and -20.78 for cases with positive TGLF values of 4.90 and 6.49, respectively. This motivates examining prediction signs and individual failures when selecting a model. The comparison concerns 43 structured cases and the settings tested here; it does not establish that forests or splines are generally unsuitable for TGLF.

3.2 Baseline performance and activation choices

On the larger database, the five-network ensemble outperformed the mean and linear baselines on all three outputs (Table 2). Relative to linear regression, test RMSE decreased by 59.6% for ion heat, 61.8% for electron heat, and 77.2% for particle flux. This does not establish superiority over tree models evaluated elsewhere, or isolate the benefit of averaging compared with an otherwise matched single network.

Table 2: Performance on the same 750 random test cases from `scan_db`. RMSE is in gyro-Bohm units. The test set was previously inspected; these are exploratory results for the selected GELU/ReLU model.

	RMSE			R^2		
Model	Q_i	Q_e	Γ_e	Q_i	Q_e	Γ_e
Mean	76.53	60.20	20.93	−0.009	−0.010	−0.011
Linear	38.43	29.48	14.43	0.746	0.758	0.519
Five-network ensemble	15.53	11.27	3.29	0.959	0.965	0.975

The controlled activation comparison selected GELU/ReLU, with a mean normalized validation RMSE of 0.1806, compared with 0.1887 for GELU/tanh (Table 3). The 4.3% aggregate improvement came from ion heat and particle flux; electron-heat error increased. Replacing all hidden activations with ReLU increased the aggregate error by 11.5% relative to GELU/tanh. Thus, neither a uniform activation replacement nor a single aggregate score captures every target's response. These results support the selected configuration under the fixed training setup, rather than a universal ordering of activation functions.

Table 3: Activation comparison on 750 validation cases. The first activation applies to both 64-unit layers and the second to the 8-unit layer. RMSE values are in gyro-Bohm units; S is defined in Eq. 1.

Activations	S	Q_i RMSE	Q_e RMSE	Γ_e RMSE
GELU / ReLU	0.1806	15.80	13.16	4.61
GELU / tanh	0.1887	17.62	12.76	4.83
ReLU / tanh	0.1935	18.15	12.93	4.99
ReLU / ReLU	0.2105	18.67	15.62	5.19

3.3 Training data and computational cost

The ensemble's errors decreased as the training set grew (Figure 1). Mean normalized test RMSE decreased by 50.2% between 250 and 3,496 training cases, compared with 2.9% for linear regression. The ensemble improved by a further 11.1% between 2,000 and 3,496 cases. This run provides evidence that the model still benefited from additional training examples over the tested range. It does not establish the amount of data needed to reach a specified accuracy.

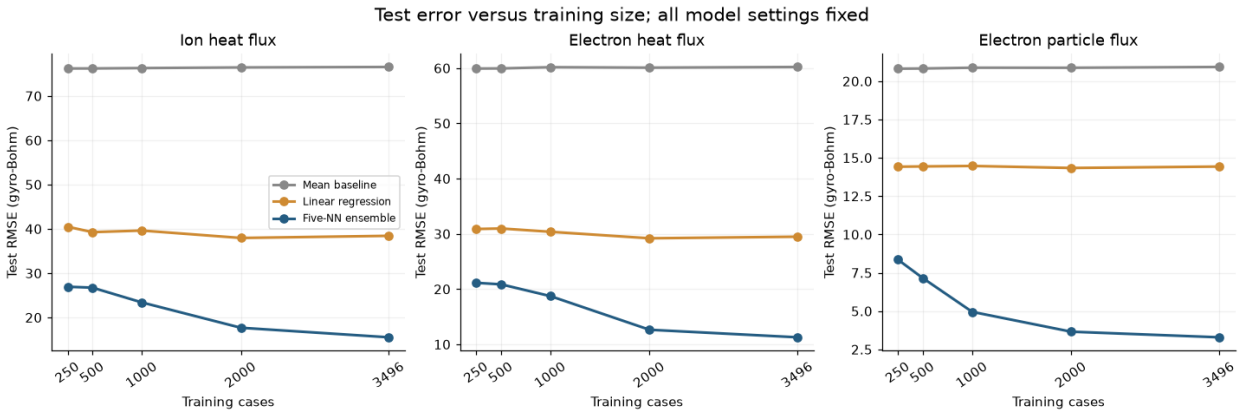


Figure 1: Learning curves for the selected GELU/ReLU five-network ensemble and both baselines. Each point uses the same previously inspected 750-case test set. Subsets form one nested sequence, with no resampling uncertainty bands. Settings are fixed, but larger subsets entail more minibatch updates per epoch.

In the recorded DSMLP CPU evaluation, training all five networks took 24.01 s. Median prediction latency was 0.814 ms for one case, while a batch of 750 averaged 4.51 μs per case. Training used four threads and prediction used one. Prediction timings used seven repetitions after warmup and included scaling, inverse transformations, and averaging, but excluded model loading, file access, and feature construction. The supplied TGLF logs do

not provide elapsed runtimes, so these measurements do not support a measured TGLF-to-surrogate speedup ratio.

3.4 Behavior beyond aggregate error

On the 43 starter cases, the selected ensemble achieved R^2 values of 0.9702, 0.9973, and 0.9876 for Q_i, Q_e, and Γ_e, respectively. Nevertheless, it matched only 25 of 27 temperature-gradient directions and 45 of 48 isotope directions, and predicted the wrong particle-flux sign in two cases. The validation-selected activation also improved starter-set R^2 for electron heat relative to GELU/tanh while reducing it for ion heat and particle flux. These checks show why an overall score should be read alongside local trends and failure cases.

The separate 15-network uncertainty experiment reached empirical random-test coverage of 91.2%, 90.1%, and 91.9% for nominal 90% intervals on the three outputs. Coverage and network agreement were less informative in some other conditions. In the outside-range collisionality group, ion-heat RMSE was 1.324 times the ordinary test value while member standard deviation was only 0.343 times as large. On the same 300 DIII-D input cases, mean predicted particle flux was 0.995 compared with TGLF's 0.483, despite 100% interval coverage. The average interval width was 3.85, in gyro-Bohm units. Broad intervals can therefore contain the reference values while the central prediction remains biased. The cause of this local bias has not been established.

4 Discussion and remaining work

These experiments show that representation and training choices can matter as much as model complexity. Logarithmic collision frequency improved the small-data linear fits. The larger-data ensemble improved on its baselines, but activation changes traded accuracy across outputs. Sign errors, incorrect trends, and local bias show why average error and interval coverage alone are insufficient.

The conclusions remain limited by the small structured starter set, exploratory model-family selection, prior test inspection, and reuse of validation data for uncertainty calibration. The learning curves cover one nested sequence. Because the experiments use different datasets and procedures, they cannot establish one model ranking or determine whether ensembles outperform the group's boosted trees or a matched single network.

4.1 Further model comparisons

We intend to test whether expanding the ensemble improves prediction enough to justify the additional training and inference cost. The 32-network model described by Meneghini et al. (2017) motivates studying member count as a design choice. A controlled comparison will hold the architecture, training data, and fitting procedure fixed while varying the number

of members, including a single-network reference. Our existing five- and fifteen-network analyses do not isolate this effect.

Other proposed experiments concern alternative transformations of the input features, different network depths and widths, and separate ensembles for Q_i, Q_e, and Γ_e in place of the current shared three-output networks. The input-transform comparisons will extend beyond the current logarithmic collision-frequency representation. Learned transformations will be fitted on training data only. These choices will be selected using validation data and evaluated for each target, with computational cost reported for the full predictor. We will consolidate the group's other models under the same evaluation procedure and preserve new assessment cases from further tuning.

4.2 Planned DIII-D study

Our mentor's November 2 project plan provides a sequence for applying these methods to a DIII-D database. The first stage is to run TGLF at 13 supplied radial locations and verify the results against the reference answers. The Task 4 handout then specifies 500–1,000 additional cases per team member, spread across these radii. Five inputs are varied around each reference condition: the two temperature gradients, shared density gradient, temperature ratio, and collision frequency. The database will retain the prescribed geometry at each radius and apply the stated cleaning rule for negative heat-flux cases.

The next stages are to train a baseline and neural network for the three fluxes, attach uncertainty estimates, and evaluate new input cases supplied by the mentors. Learning curves using 500, 1,000, 2,000, and all available training cases will examine whether errors continue to decrease, intervals shrink, and some input regions remain poorly represented. The final comparison will evaluate the broad `scan_db` surrogate and the DIII-D-trained surrogate on the same DIII-D test cases, assessing both accuracy and the reliability of uncertainty estimates.

The plan also lists two possible extensions: comparing surrogate error with the discrepancy between TGLF and nonlinear CGYRO outputs, and examining the contributions of low- and high-wavenumber turbulence to electron heat flux. These extensions are separate from the main comparison. Throughout the study, the plan emphasizes correlated plasma inputs, data quality, uncertainty estimates for every prediction, and flags for inputs outside the training data. This work remains distinct from the completed experiments reported above.

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